A negative-cable counterexample to Problem 00000000614

Actual diagrams and filtered rational Lee homology

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Claim and scope. The printed formula, with no restriction on the sign of q, is

$$s(C_{p,q}(K)) = p\, s(K) + (p-1)(q-1).$$

Take the crossingless unknot U and $(p,q) = (2,-1)$. We compute both invariants from their actual unreduced Lee complexes over $\mathbb{Q}$:

$$s(C_{2,-1}(U)) = s(U) = 0, \qquad 2 \cdot 0 + (2-1)(-1-1) = -2.$$

Here $p > 0$ and $\gcd(2,-1) = 1$. This refutes only the unrestricted printed formula. It does not refute a version with positive q or another framing convention.

Published geometric boundary. We use the standard zero-framed satellite convention of Lewark–Zibrowius, Definition 3.13: $t \in \mathbb{Z}$ and $C_{2,2t+1}(U) = T_{2,2t+1}$; see printed pages 259–260. Thus $t = -1$ is allowed. Identification of the product solid torus with the zero-framed tubular neighbourhood of the unknot is this published geometric identification. Lean does not reconstruct general three-manifold topology, satellite invariance, Reidemeister invariance, or the Lee spectral sequence. No such fact is an added Lean axiom. The particular torus curve, its entire two-sheet braid, signs, finite incidence and resolutions are constructed and checked in Lean.

The actual negative diagram. On the genuine unit circle, use the embedded curve

$$\gamma : \mathbb{T} \longrightarrow \mathbb{T} \times \mathbb{T}, \qquad z \longmapsto (z^2, z^{-1}).$$

It lies on the boundary of the product solid torus, is injective, and is disjoint from the core $z \mapsto (z,0)$. Over longitude $e^{2\pi i t}$, $0 \le t \le 1$, its two sheets have meridians $u_0(t) = e^{-\pi i t}$ and $u_1(t) = -e^{-\pi i t}$. Lean proves these are all fibres, that they cover the entire curve, and that their endpoints interchange.

Project to $(\Re u_s(t), -t)$, retaining $\Im u_s(t)$ as depth. There is exactly one double point, $t = 1/2$. Sheet 1 is above sheet 0. The actual over/under tangents are $(\pi,-1)$ and $(-\pi,-1)$, whose oriented crossing determinant is $-2\pi < 0$. Therefore $n_+ = 0, n_- = 1$. The four ports are top left, top right, bottom left, bottom right, numbered $0,1,2,3$. The strand involution is $(03)(12)$, the closure is $(02)(13)$, and the clockwise rotation is (0132). Their analytic endpoint and tangent identifications are proved. The closed graph is connected. In Rasmussen's Figure 1 conventions, the 0-resolution is $(01)(23)$ and the 1-resolution is $(02)(13)$. Actual graph component quotients give respectively one and two circles, hence a split saddle.

The rational Lee algebra and full complex. The native algebra is $A = \mathbb{Q}[\omega]/(\omega^2 - 1)$, implemented as `QuadraticAlgebra`; $v_+ = 1$, $v_- = \omega$ form its native basis. Actual multiplication and tensor comultiplication satisfy Rasmussen's equations (3):

$$1 \cdot 1 = \omega \cdot \omega = 1, \quad 1 \cdot \omega = \omega \cdot 1 = \omega, \qquad \Delta(1) = 1 \otimes \omega + \omega \otimes 1, \quad \Delta(\omega) = \omega \otimes \omega + 1 \otimes 1.$$

The actual circle-indexed tensor powers are linearly equivalent to A and $A \otimes A$. Under these proved equivalences, the geometric saddle is Δ. The one-edge cube sign is $+1$. The full native complex, indexed by every $i \in \mathbb{Z}$, is

$$C^{-1} = A \xrightarrow{\Delta} C^0 = A \otimes A, \qquad C^i = 0 \quad (i \neq -1, 0).$$

It is a genuine `CochainComplex (ModuleCat Q) Z`, with $d^2 = 0$. Injectivity of Δ is proved, so native homology is zero outside degree 0. A canonical native homology isomorphism identifies

$$H^0(C) \simeq (A \otimes A)/\operatorname{im}\Delta.$$

Both normalization shifts are retained. For an enhanced state, Rasmussen uses $\mathrm{gr} = |v| - n_-$ and $q = p + \mathrm{gr} + n_+ - n_-$. Thus the negative diagram's output has $q = p - 1$, not p:

C^{-1} basis	1	ω
q	-1	-3

C^0 basis	00	01	10	11
q	1	-1	-1	-3.

These degrees follow from actual circle labels and actual crossing counts. For every integer j, the cochain filtration consists of vectors whose coefficients of basis vectors with $q < j$ vanish. Lean proves Δ preserves it. The homology filtration $F^j H$ is the image of its actual filtered cycle representatives, transported through the canonical native homology isomorphism.

An explicit quotient calculation, not assigned homology values. Write a tensor vector as $x = (x_{00}, x_{01}, x_{10}, x_{11})$. Then

$$\Delta(a + b\omega) = (b, a, a, b), \qquad P(x) = (x_{00} - x_{11}, x_{01} - x_{10}).$$

The true equalities $\ker P = \operatorname{im}\Delta$ and surjectivity of P are proved. The section is $(a, b) \mapsto a(00) + b(01)$. Consequently the native quotient has a proved coordinate isomorphism to $\mathbb{Q}^2$; write $\overline{P}$ for this induced map. For every $j \in \mathbb{Z}$, its actual filtered image is exactly

$$\overline{P}(F^j H) = \{(a, b) : \ j > 1 \Rightarrow a = 0, \quad j > -1 \Rightarrow b = 0\}.$$

The forward inclusion follows from all four actual tensor degrees; the reverse inclusion uses the displayed genuine section. In particular, the low-degree 11 representative does not force its class to have degree -3: the same class has the representative -00 of degree 1.

Actual extrema and the crossingless companion. For a nonzero native homology class h, define its filtration degree by

$$d(h) = \sup\{j \in \mathbb{Z} : h \in F^j H\}.$$

Lean proves each such set is nonempty and bounded above. In the actual coordinates (a, b) it is $(-\infty, -1]$ if $b \neq 0$, and $(-\infty, 1]$ if $b = 0$ and $h \neq 0$. Actual native classes with coordinates $(0, 1)$ and $(1, 0)$ attain both endpoints. Thus the set of nonzero-class degrees is exactly $\{-1, 1\}$; its minimum and maximum are genuine attained extrema, not empty-set or unbounded default values.

The companion is the actual planar unit circle. Its projection is injective and its native topological connected-component type has one element. Its circle tensor space is proved equivalent to A. Its full integer complex is A in degree 0, zero elsewhere, with zero differential. Native homology is A, with basis degrees $1, -1$, and all other native homology groups are zero. The image-of-cycles filtration obeys the same all-integer coordinate formula. Hence both diagrams have

$$s_{\min} = -1, \qquad s_{\max} = 1, \qquad s = s_{\max} - 1 = s_{\min} + 1 = 0$$

in Definitions 3.1 and 3.4 of Rasmussen. This proves the stated counterexample without assuming a precomputed unknot invariant.

Formalization and reproducibility. The complete 969-line `lean/Main.lean` has 175 explicitly named declarations, plus three anonymous instance commands. The dynamic `Check.lean` enumerates every owned name, including generated and anonymous instance names, and runs ordinary `#print axioms` for each. Only `propext`, `Classical.choice`, and `Quot.sound` are allowed; there are no extra axioms, placeholders, or `native_decide` proofs.

The pins are Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. Run `lean --version`, then `bash lean/verify.sh` from the appropriate project directory for build, warning-as-error declaration audit, and the official fresh checker. The submission records the four individual commands, source hashes and original resource observations. Production elapsed times are diagnostic; they establish no controlled research-efficiency improvement.

Primary references actually used.

1. J. Rasmussen, *Khovanov homology and the slice genus*, arXiv:math/0402131v1 (9 February 2004), printed pp.3–4 and 6: the resolutions, normalized cube, rational Lee maps and filtered homology definitions. `https://arxiv.org/abs/math/0402131`. The un-oriented curl on p.15 is not used to assign this braid's sign.

2. L. Lewark and C. Zibrowius, *Rasmussen invariants of Whitehead doubles and other satellites*, J. reine angew. Math.816 (2024),241–296, printed pp.259–260, Definition3.13 and its satellite convention. `https://doi.org/10.1515/crelle-2024-0061`. Only its geometric convention is used, not its characteristic-2 invariant.